

MINI PROJECT – BEGINNER DATA PIPELINE

STEP 1 – Create Project Folder (Linux)

Open terminal.

Create a project folder:

student_project

Inside it create the following files:

students.csv, clean_students.csv, analysis.sql

```
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice$ mkdir -p student_project
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice$ cd student_project/
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/student_project$ touch students.csv clean_students.csv analysis.sql
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/student_project$ ls -lrth
total 0
-rwxrwxrwx 1 prashant prashant 0 Mar  9 2026 students.csv
-rwxrwxrwx 1 prashant prashant 0 Mar  9 2026 clean_students.csv
-rwxrwxrwx 1 prashant prashant 0 Mar  9 2026 analysis.sql
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/student_project$ |
```

STEP 2 – Create Sample Dataset

Open students.csv and add sample data.

```
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/student_project$ cat students.csv
id,name,age,marks,city
1,Rahul,20,85,Mumbai
2,Sneha,21,78,Pune
3,Amit,22,90,Delhi
4,Priya,20,65,Nashik
5,Rohit,23,88,Mumbai
6,Anjali,21,72,Pune
7,Karan,22,95,Delhi
8,Meena,20,60,Nashik
9,Vijay,24,81,Mumbai
10,Pooja,21,75,Pune
11,Arun,23,69,Delhi
12,Neha,22,84,Nashik
13,Sanjay,24,77,Mumbai
14,Kavita,20,91,Pune
15,Deepak,23,73,Delhi
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/student_project$ cat students.csv | column -t -s,
id  name  age  marks  city
1   Rahul  20   85     Mumbai
2   Sneha  21   78     Pune
3   Amit   22   90     Delhi
4   Priya  20   65     Nashik
5   Rohit  23   88     Mumbai
6   Anjali 21   72     Pune
7   Karan  22   95     Delhi
8   Meena  20   60     Nashik
9   Vijay  24   81     Mumbai
10  Pooja  21   75     Pune
11  Arun   23   69     Delhi
12  Neha   22   84     Nashik
13  Sanjay 24   77     Mumbai
14  Kavita 20   91     Pune
15  Deepak 23   73     Delhi
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/student_project$ |
```

STEP 3 – Python Task

Write a Python program called
process_students.py

Tasks:

1. Read students.csv file.
2. Remove students whose marks are less than 70.
3. Convert names to uppercase.
4. Save the cleaned data into a new file:

clean_students.csv

```
'''  
  
# 1.Read students.csv file.  
import csv  
  
with open("students.csv", "r") as file:  
    data = csv.reader(file)  
    for row in data:  
        print(row)
```

```
# 2.Remove students whose marks are less than 70.  
  
clean_students_data = []  
  
with open("students.csv", "r") as file:  
    data = csv.DictReader(file)  
    for row in data:  
        if int(row['marks']) > 70:  
            marks = row.copy()  
            marks['name'] = marks['name'].upper()  
            clean_students_data.append(marks)  
print(clean_students_data)  
  
with open("clean_students.csv", "w") as file:  
    writer = csv.DictWriter(file, fieldnames= ['id', 'name', 'age', 'marks', 'city'])  
    writer.writeheader()  
    writer.writerows(clean_students_data)  
  
with open("clean_students.csv", "r") as file:  
    data = csv.DictReader(file)  
    for row in data:  
        print(row)
```

STEP 4 – SQL Task

Create a table called students.

Columns : id, name, age, marks, city

Insert the cleaned data into the table.

Write SQL queries to answer:

1. Show all students.
2. Find average marks.
3. Find student with highest marks.
4. Count number of students from each city.

```
# STEP 4 - SQL Task
'''
CREATE TABLE STUDENT(id NUMBER, name VARCHAR2(20), age NUMBER, marks NUMBER, city VARCHAR2(20));
INSERT INTO STUDENT VALUES (1, 'Rahul', 20, 85, 'Mumbai');
INSERT INTO STUDENT VALUES (2, 'Sneha', 21, 78, 'Pune');
INSERT INTO STUDENT VALUES (3, 'Amit', 22, 90, 'Delhi');
INSERT INTO STUDENT VALUES (4, 'Priya', 20, 65, 'Nashik');
INSERT INTO STUDENT VALUES (5, 'Rohit', 23, 88, 'Mumbai');
INSERT INTO STUDENT VALUES (6, 'Anjali', 21, 72, 'Pune');
INSERT INTO STUDENT VALUES (7, 'Karan', 22, 95, 'Delhi');
INSERT INTO STUDENT VALUES (8, 'Meena', 20, 60, 'Nashik');
INSERT INTO STUDENT VALUES (9, 'Vijay', 24, 81, 'Mumbai');
INSERT INTO STUDENT VALUES (10, 'Pooja', 21, 75, 'Pune');
SELECT * FROM STUDENT;

-- 1. Show all students.
SELECT * FROM STUDENT;

--2. Find average marks.
SELECT AVG(MARKS) AS AVG_MARKS FROM STUDENT;

-- 3. Find student with highest marks.
SELECT ID,NAME,MARKS FROM STUDENT ORDER BY MARKS DESC FETCH FIRST 1 ROW ONLY;

-- 4. Count number of students from each city.
SELECT DISTINCT CITY FROM STUDENT;
SELECT CITY, COUNT(*) AS COUNT_CITY FROM STUDENT GROUP BY CITY;
'''
```

STEP 5 – Linux Task

Use Linux commands to:

1. Count number of lines in students.csv.
2. Search students from city Pune.
3. Display first 3 lines of the file.

Commands to explore

wc, grep, head

```
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/linux_exercise/data_lab$ cat students.csv | wc -l
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prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/linux_exercise/data_lab$ grep -i "Pune" students.csv
2,Sneha,21,78,Pune
6,Anjali,21,72,Pune
10,Pooja,21,75,Pune
14,Kavita,20,91,Pune
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/linux_exercise/data_lab$ head -n 3 students.csv | column -t -s,
id  name  age  marks  city
1   Rahul  20   85     Mumbai
2   Sneha  21   78     Pune
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/linux_exercise/data_lab$ |
```